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Application to the ETH Zurich Ethics Commission

Version *ETHZ_Ethics_Application_V2025* (newest version available here)

Megastudy testing 20 information interventions to strengthen trust in climate scientists in the US

Principal Investigator (PI)[1]

Name	Title	Group/Chair/Institute	University
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Involved Researchers

Name	Title	Group/Chair/Institute	University
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Hugo Mercier	Prof.	Institut Jean Nicod	CNRS
Naomi Oreskes	Prof.	Department of the History of Science	Harvard University
Mario Scharfbillig	Prof.	Joint Research Centre (JRC), European Commission	
Mike S. Schäfer	Prof.	Department of Media and Communication	University of Zurich

Overview

Project type (Check all that apply)	☐ Bachelor's thesis ☐ Master's thesis ☐ Doctoral thesis ☒ Research project Name of student:

[1]

Further Information

The principal investigator (PI) or Co-PI must be affiliated with ETH Zurich (student, employee). The PI of regular research projects is usually a professor or a senior scientist with budget responsibility. If students are primarily responsible for the project, they should be indicated as PI.

	Bachelor/Master applications: I, *name*, confirm that my supervisor *name* has reviewed this application
Method(s) of data collection (Check all that apply)	☐ Interview (☐ in person ☐ phone ☐ online) ☐ Focus group (☐ in person ☐ online) ☒ Survey (☐ in person ☐ phone ☒ online) ☒ Experimental behavioural study (☐ in person ☒ online) ☐ Social media (☐ observation ☐ intervention) ☐ Incomplete information or deception of participants ☐ Physiological measurements ☐ Mobile app (☐ incl. tracking) ☐ Photo-/video-/audio recording ☐ Secondary analysis of personal data ☐ Data from lectures/students[2] ☐ Other methods:
Number of participants	Minimum: 22,000 (1,000 participants per treatment condition, 2'000 for the control condition) Maximum: 22,000
Brief description of participants	Participants will be recruited from a national, non-probability panel of American residents provided by CloudResearch. The sample will be quota-matched to the U.S. population for several demographic variables, including gender, age, race/ethnicity, education, and region. We will only recruit adult participants who are more than 18 years old.
Age group of participants	☒ 18 Years or above ☐ between … and … years old

1. Abstract

Trust in climate scientists has been found to be the strongest predictor of belief in climate change and support for policies to mitigate climate change. Strengthening trust in climate scientists may therefore be an important lever for accelerating climate action. However, there is currently little evidence on how to strengthen trust in climate scientists. We aim to address this gap by systematically identifying, testing, and comparing information interventions designed to strengthen trust in climate scientists. With a sample of N = 22,000 US participants, we experimentally test 20 text-based interventions identified through a crowdsourcing process and test their effect on trust in climate scientists—our main outcome variable. We will also assess the interventions' impact on various other outcome variables, including, for instance, belief in climate change, support for climate policies and public research funding, and taking individual action to foster climate research. We do not foresee any risks. The results of our study have the potential to inform the communication strategies of a wide range of actors—e.g., universities, research institutes, governments, or NGOs—looking to rally support in the fight against climate change.

2. Project

2.1 Study Objective

Our primary research question is: Which information interventions significantly increase trust in climate scientists? There are three groups of secondary research questions: First, how do the information interventions affect other outcomes of interest? Second, are the effects of the information interventions moderated by other variables? Third, do the information intervention effects persist, such that they can still be detected a week later? We hypothesize that our interventions will significantly increase trust in climate scientists compared to the control group.

Trust in climate scientists plays a critical role in addressing climate change. Trust in climate scientists has been shown to relate to public acceptance, support, and adherence to science-based policy

[2] Please take note of the guideline on research in an educational setting.

measures. Given the urgent need to scale up action on climate change, increasing trust in climate scientists may be an important lever to accelerate climate action and strengthen societal resilience to climate change. Despite the importance of trust in climate scientists, research on behavioral interventions to enhance it remains scarce, highlighting its potential as a key intervention target. The project therefore addresses a critical research gap by identifying effective interventions to increase trust in climate scientists.

2.2 Methods and Study Design

After providing their informed consent to take part in the study, participants are informed that they must qualify for the study. They complete a demographic questionnaire, including questions on gender, age, race, education, income, social class, their residential zone, partisan identity and religion. Among the demographic questions, participants need to pass a first attention check, which, if they fail, results in immediate exclusion from the study. Participants who pass are informed that they qualify for the study, that the study is about their opinion about climate change and climate scientists, and a definition of climate scientists ("Climate scientists study changes in the Earth's climate over time and how they might affect the planet in the future."). Next, participants answer questions on a set of pre-treatment variables, including belief in climate change, trust in climate scientists (single-item measure), perceived alienation from climate science and need for epistemic autonomy. Next, participants respond to a second attention check, but failing this second test does not result in exclusion from the rest of the study.

Participants will then be randomly assigned to one of 21 experimental conditions (20 intervention conditions and one control condition). In each of the intervention conditions, participants will read a short text-based informational intervention about climate science or climate scientists. In the control condition, participants will be randomly assigned to read one of three neutral texts unrelated to climate change. These texts cover (a) the history of neckties, (b) the rules of baseball, and (c) different types of dances. Using multiple control texts reduces the risk that unintended characteristics of any single text influence the outcome variables.

After being exposed to the content of their respective conditions, participants will complete a set of outcome measures. There are three groups of outcomes. The primary outcome is a multidimensional measure of trust in climate scientists. Secondary outcomes assess attitudes directly related to climate science and scientists, views on the role of climate scientists in policymaking, trust in public climate research institutions, and trust and distrust in climate scientists (single-item measures). In addition, the block of secondary outcomes contains two behavioral measures: For one, participants are asked to allocate money between themselves and the American Geophysical Union (AGU). For the other, participants are given the opportunity to subscribe to the free version of a climate scientist's newsletter (Katharine Hayhoe's "Talking Climate" newsletter).

The third block of tertiary outcomes consists of variables related more generally to climate change and climate change mitigation, including belief in and concern about climate change, support for climate change mitigation policies, and individual mitigation behaviors.

Approximately one week after the main experiment, a follow-up survey will be conducted to assess the persistence of the most effective interventions. Participants from the five intervention conditions that show the strongest effects, as well as participants from the control group, will be invited to complete the outcome measures again.

No deception or incomplete disclosure of information will take place.

Survey participants will receive detailed instructions on how to complete the questionnaire.

2.3 Participants

Recruitment process and number of participants: Participants will be recruited from a national, non-probability panel of American residents provided by CloudResearch. The sample will be quota-matched to the U.S. population for several demographic variables, including gender, age, race/ethnicity, education, and region. We will only recruit adult participants who are more than 18 years old. Data collection will be stopped as soon as we have collected complete responses from 22,000 participants (1,000 participants for each of the 20 treatment conditions, 2'000 participants for the control condition). Before treatment assignment, we will deploy a series of attention and bot detection checks. For some of these tests, only those participants who have passed will be able to continue and complete the study.

Sample size calculation: According to Monte Carlo power simulations, a sample size of N = 22,000 would allow us to detect our smallest effect of interest, Cohen's d = 0.2, with statistical power of nearly 100%. An effect size of Cohen's d = 0.2 is typically considered small in behavioral sciences. In a pilot sample (N = 76), this minimal effect size of interest would translate to a 3-point change of the sample mean on a scale from 0, very low trust, to 100, very high trust. In the simulations, we adjusted p-values for multiple testing via the Benjamini–Hochberg false discovery rate (FDR) procedure and used $\alpha = .05$ as the cutoff point for statistical significance.

Exclusion: First, we do not allow for any individual participant taking part in our study several times. In cases with a duplicated participant ID, we will only keep the first case. Second, we will exclude participants who failed a series of attention and bot detection checks. These checks will be run before treatment assignment to avoid post-treatment bias (Montgomery et al., 2018).

Montgomery, Jacob M., Brendan Nyhan, and Michelle Torres. 2018. "How Conditioning on Posttreatment Variables Can Ruin Your Experiment and What to Do about It." *American Journal of Political Science* 62 (3): 760–75. https://doi.org/10.1111/ajps.12357.

2.4 Timeline

	February 2026	March 2026	April 2026	May 2026
Survey design				
Data collection				
Data analysis & Writing				

2.5 Partners and Funding

The study will be funded by the Stanford University Start-Up Funds of Prof. Madalina Vlasceanu and Prof. Sara Constantino. Project partners do not have any conflict of interest.

3. Ethical Aspects

3.1 Informed Consent and Debriefing

Please find the consent form in the appendix. Please also note that this online study will be run in collaboration with Stanford University in the United States (US), who also funds data collection. The Stanford IRB is currently reviewing our application for this project. As data collection will take place in the US, we have attached the consent form in line with Stanford IRB's guidelines, knowing that they might differ slightly from ETH requirements.

3.2 Data Protection and Publication

No personal data will be collected in this study. Please note that we will ask participants to provide their postcodes, but participants cannot be identified based on the data collected in our survey. Anonymous data will be published on the Eawag Research Data Institutional Collection as well as on the Open Science Framework. The results will be published in peer-reviewed journals.

3.3 Compensation

Participants receive monetary compensation for participating in the study. They will receive the amount they agreed upon with the data collection company CloudResearch.

3.4 Risks and Discomforts

We do not expect any social or ecological risks from this survey. Participants can end their participation at any time and data is collected anonymously.

4. References

Appendix 1: Information and consent documents for participants

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Information and consent form

You are invited to participate in a research study that will ask about your opinions and attitudes. You must be at least 18 years of age to participate. There are no risks associated with this study and your identity will be kept confidential. We cannot and do not guarantee or promise that you will receive any benefits from this study.

Participation

If you decide to participate in this project, please understand your participation is voluntary and you may withdraw your consent or discontinue participation at any time without penalty. The alternative is not to participate. You have the right to refuse to answer particular questions. Your individual privacy will be maintained in all published and written data resulting from the study.

Contact Information

If you have any questions, concerns or complaints about this research, its procedures, risks and benefits, contact the Protocol Director, Madalina Vlasceanu at mov@stanford.edu. If you wish to contact someone independent of the researchers, you may email the Stanford Institutional Review Board at irbnonmed@stanford.edu.

If you agree to participate in this research, please click to the next screen and complete the questionnaire.

Appendix 2: Data collection tools (surveys, questionnaires, etc.)